

A-level

ENVIRONMENTAL SCIENCE

Paper 1 - Generated practice paper Set 06

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

- Marking points are separated by bullets; each bullet is worth 1 mark unless stated.
- Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
- Alternative correct wording and other relevant, accurate points are credited.
- For levels-of-response questions, first determine the level using the descriptors, then position the mark within the level using the indicative content.
- Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Control of greenhouse gases: energy conservation (transport, buildings, industry), renewable and nuclear energy replacing fossil fuels, CCS, reduced deforestation and afforestation (carbon sequestration), methane capture from landfill / digesters, HFC controls, changes in agriculture • Geoengineering: white roofs, ocean fertilisation, solar shades - largely untried, possible unpredictable consequences • International agreements: Kyoto Protocol, Paris Agreement - targets, but non-binding / uneven participation; carbon trading and taxes; comparison of per capita carbon footprints between countries • Evidence of limited success: global CO₂ concentrations still rising (about 280 ppm pre-industrial to over 410 ppm); temperatures up about 1 °C; emissions in some developed countries falling but rising in developing economies • Reasons for difficulty: fossil fuels underpin economies, need for major lifestyle change, cost, time delays and natural fluctuations masking effects, political disagreement, developing countries' right to develop • Adaptation strategies (flood defences, drainage, managed retreat, infrastructure changes) reduce impacts but do not address the cause • Comparison with the Montreal Protocol's success for ozone • Judgement: partial success - technology exists and some regional reductions achieved, but global strategies have not yet stopped the rise; tipping points make urgency greater	9
01.2	• The total greenhouse gas / CO₂ emissions caused by an individual, organisation, product or country (often expressed per capita per year)	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 89-107; AQA spec 3.2.1.2.	
02.1	• 15 - (-18) = 33 °C	1
02.2	• Visible light passes easily through the atmosphere and is absorbed by the surface, which warms • The warm surface emits (long-wave / far) infrared radiation • Greenhouse gases (water vapour, CO₂, methane) absorb infrared better than the average for the atmosphere and convert it to heat • The warm atmosphere emits infrared which is absorbed by the surface, and it reduces heat loss from land and oceans by conduction	4
02.3	• Any gas that is better at absorbing infrared radiation than the average for the atmosphere	1

Question	Marking guidance	Marks
02.4	• Arriving: mainly ultraviolet, visible light and near infrared (short wavelength); leaving: mainly long-wavelength (far) infrared • Gases that are transparent to the incoming short wavelengths can absorb the outgoing long wavelengths, so changing their concentration changes how much energy is retained	2
02.5	• The energy and the processes it drives control climate, ocean currents and the hydrological cycle • These determine the abiotic conditions (temperature, rainfall) that each species is adapted to, so a change alters where species can survive	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 87-89; AQA spec 3.2.1.1.	
03.1	• Aerobic decomposition produces little methane and returns nutrients to the soil	1
03.2	• It may contain concentrated heavy metals / toxic residues • It is stored in solid form at high pH / in hazardous-waste landfill so the metals stay insoluble	2
03.3	<i>Any 2 from:</i> • Impermeable polymer / clay liner to stop leachate escaping • Collection and treatment of leachate • Collection of methane and use as fuel • Daily covering of waste / final capping and restoration	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 279-285; AQA spec 3.4.3.2.12.	

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04.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Neonicotinoids are neurotoxins similar to nicotine; they inhibit the neurotransmitter acetylcholine • High toxicity to insects but low toxicity to vertebrates because they cannot cross the vertebrate blood-brain barrier • Very toxic to bees: sub-lethal doses reduce navigation ability, foraging and immunity to disease; linked to colony decline • Water soluble and moderately persistent: remain in soil, absorbed by roots and translocated (systemic), so present in pollen and nectar • Persistence may allow build-up in aquifers / soil; broken down fastest in sunlight and by soil microbes • Synergism with some fungicides increases toxicity to bees • Exposure greatest when sprayed onto crops or when seed-coating dust is blown by wind • Control: seed coating rather than spraying; spraying when crops are not in flower / at night; still days to reduce drift • Restrictions on use (EU bans on outdoor use on flowering crops); use of alternative pest control (cultural, biological, integrated control) • Evaluation: reduced pesticide use but neonicotinoid alternatives (pyrethroids) also harm pollinators; trade-off with crop yields	9
04.2	<i>Any 1 from:</i> • Biological control (introduced predators / pathogens) • Predator habitats (hedgerows, beetle banks) • Crop rotation • Companion / barrier crops • Sterile male technique / pheromone traps • Genetic resistance	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 265-266; AQA spec 3.4.3.2.8.	
05.1	• Ground surface: pitfall trap (container sunk level with the surface, cover, checked after 24 h) - only mobile surface animals; trapped predators eat others; preservative attracts / repels; escapes • Leaf litter: Tullgren funnel (sample on mesh under a light; heat / light drive invertebrates down into a container for a standard time) - only species repelled by heat / light; some die or stop moving; large ones cannot pass the mesh • Branches: beating tray (branch beaten a set number of times over a sheet) - hard to standardise vigour; some species not dislodged / fly away; high branches unreachable	6

Question	Marking guidance	Marks
05.2	• Moths only fly as adults for part of the year and activity depends on weather / season, so a species may be present but not caught • Brightness / wavelength of the lamp affects attractiveness so catches vary between traps	2
05.3	• The area of ground covered by the suction pipe is known so invertebrates per m² can be calculated • A pooter is only a handling / collection device for individuals already found, not a sampling method	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 397-399; AQA spec 3.7.2.1.	
06.1	• $4.2 / 0.7 = 6$	1
06.2	• Long summer days (almost 24 hours of daylight) so the total energy received is high, even though the Sun is low • In winter days are very short / the Sun does not rise, so light levels are extremely low	2
06.3	• Reflectors can only focus parallel rays of direct sunlight onto the absorber • Clouds, smoke and dust scatter the light so it cannot be focused and intensified	2
06.4	• Very large areas of collectors are needed to harness significant amounts of energy (and the changing angle of incidence reduces it further)	1

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06.5	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Problems: intermittency (daily and seasonal cycles), unpredictable cloud cover, low energy density, changing angle of incidence, land requirements, water for cleaning in deserts, toxic wastes in manufacture (silicon tetrachloride, cadmium), efficiency loss at high temperature • Photothermal: evacuated tubes reduce conduction / convection losses; thermal stores (water, sand, concrete, molten salt) allow later use; passive solar architecture, brise soleil, light tubes; heat pumps deliver up to four times the energy used • Photovoltaic developments: multijunction cells absorbing more wavelengths, anti-reflective / moth-eye textured surfaces (smooth cells reflect 30%), PV-thermal hybrids cooling the cell, transparent PV in windows, heliostats keeping the optimum angle, self-cleaning nanohydrophobic surfaces • Concentrating solar power with molten-salt storage to 550 °C so steam turbines run when needed - overcomes intermittency • Integration: grid-connected solar farms, rooftop space avoiding land conflicts, batteries, pumped storage, links to energy storage and demand management • Evaluation: strong in sunny deserts / low latitudes, weaker in high latitudes in winter; cost falling; manufacturing impacts; no single technology solves all problems; justified conclusion	9
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 199-203; AQA spec 3.3.4.1.	
07.1	• B - tiny pores between clay particles allow capillary rise; none in sand	1
07.2	• Hygroscopic: held tightly by adhesive forces on particle surfaces; cannot drain and cannot be absorbed by plants • Capillary: held in pores by surface tension; does not drain but can be absorbed by plants • Gravitational: drains freely through large pores; available to plants only briefly as it passes	3
07.3	• Small round crumbs leave spaces giving good drainage, aeration and easy root penetration; large flat platy peds pack together reducing all three	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 158-169; AQA spec 3.2.5.	
08.1	• $(4.8 - 1.4) / 4.8 \times 100$ • = 70.8% (accept 71%)	2

Question	Marking guidance	Marks
08.2	• A tributary joins the river, more than doubling the flow ($15 \text{ to } 36 \text{ m}^3 \text{ s}^{-1}$) • So the pollutant is diluted by the extra water (dispersal), not broken down	2
08.3	• The pollutant is persistent so it is not degraded / broken down as it travels • Little extra water joins the river (flow 36 to 38) so little further dilution	2
08.4	<i>Any 2 from:</i> • Lake has a long residence time / water does not flow away so the pollutant accumulates rather than being carried away • Smaller volume of water / less dilution / less dispersal by currents • Persistent pollutant gradually becomes more concentrated (eg by evaporation)	2
08.5	• Marine discharge: large volume of water and currents dilute and disperse the pollutant to acceptably low concentrations (control of emission location) • River: small volume so concentrations stay high / drinking water abstraction / sensitive wildlife downstream	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 236-241, 254-255; AQA spec 3.4.1, 3.4.2.1.	
09.1	<i>Any 3 from:</i> • Stress-related illness: ulcers, high blood pressure, heart disease • Behavioural changes: irritability, aggression, lack of concentration leading to accidents • Communication problems / educational underperformance (eg schools under flight paths) • Sleep disturbance	3
09.2	• Acoustic fatigue: objects exposed to sound at their natural resonant frequency vibrate, causing stress cracking / structural failure (eg early jet engine cowlings) • Vibration damage: repetitive vibration from heavy vehicles damages buildings, bridges, underground pipes • Shock impacts: the force of a sudden very loud noise (sonic boom) breaks windows / dislodges masonry	3
09.3	• Birds that sing to establish territories / attract mates cannot be heard against background noise • Birds scared by sudden noise leave nests so predators take eggs / chicks; cliff-nesting seabirds knock eggs or chicks into the sea	2
09.4	• Whales / dolphins use hearing / echolocation to find food and communicate; loud sonar causes hearing damage if close to the source • Behavioural changes / disorientation possibly linked to mass strandings; reduced feeding success	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 287; AQA spec 3.4.3.2.13.	

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10.1	Levels of response Level 3 (7-9 marks): Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. Level 2 (4-6 marks): Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. Level 1 (1-3 marks): Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. 0 marks: No relevant content. Indicative content (credit other relevant, accurate points) • Features: renewable / non-renewable, depletable, abundance, energy density, locational constraints, intermittency and predictability, need for conversions, applicability, ease of storage and transport, technological development, political and economic influences, environmental impacts • Fossil fuels: high energy density, easy storage and transport, reach high temperatures, well-developed technologies and infrastructure - but finite, polluting (acid rain, smoke, CO₂), politically concentrated (Middle East oil) • Nuclear: very high energy density, base load, few fuel deliveries - but complex, expensive, waste, public opposition • Renewables: low energy density, intermittent (wind, solar unpredictable; tidal predictable), locationally constrained, cannot be stored without conversion, need storage / grid, cannot reach fossil-fuel temperatures or replace liquid vehicle fuels yet - but renewable, low pollution, falling costs • Sustainability concepts: impacts before use (extraction, processing, equipment embodied energy) and as a consequence of use (pollution, habitat damage, depletion) • Economic and political factors: subsidies, guaranteed prices, planning, external costs not included in price, energy security • Judgement: a diverse mix matched to uses (electricity vs heat vs transport) with storage is most sustainable; transition must start before depletion	9
10.2	• Very high energy density fuel means only a few tens of tonnes are needed each year, compared with 10 000 t of coal per day for a coal-fired station	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 175-182; AQA spec 3.3.2-3.3.3.	

Question	Marking guidance	Marks
11.1	Discuss how the atmosphere supports life on Earth and evaluate the extent to which human activities threaten these life-support functions. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Composition and structure: N₂, O₂, Ar, CO₂, ozone; troposphere and stratosphere; pressure and liquid water; gases for biological molecules and industry • Energy processes: UV absorbed by O / O₂ / O₃ dynamic equilibrium; visible light absorbed at the surface; infrared absorbed by greenhouse gases (33 °C warming); heat distribution by winds and ocean currents; transport of water vapour; energy budget and albedo • Global climate change: anthropogenic greenhouse gases, sources, feedback and tipping points, consequences for climate, sea level, currents, species; difficulties of monitoring and prediction; control strategies and agreements • Ozone depletion: CFCs and halons, catalytic destruction, effects of UV, Montreal Protocol and replacements, recovery time • Other threats: aerosols, smoke, land-use change; interconnection of processes • Evaluation: which functions are most at risk; success of ozone controls vs slow progress on greenhouse gases; justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 86-113; AQA spec 3.2.1.	25

Question	Marking guidance	Marks
11.2	Discuss the strategies that can be used to provide sustainable supplies of water for a growing population. Levels of response Level 5 (21-25 marks): Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. Level 4 (16-20 marks): Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. Level 3 (11-15 marks): Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. Level 2 (6-10 marks): Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. Level 1 (1-5 marks): Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. 0 marks: No relevant content. Indicative content (students are not expected to cover all of these; credit other relevant material) • Context: uneven distribution of water, growing population and per capita demand, unsustainable exploitation of rivers and aquifers, hydrological cycle links • New sources: rainwater collection, rivers (discharge, fluctuations, contaminants), reservoirs and site selection, estuary barrages, desalination, unexploited aquifers • Management: artificial aquifer recharge, river-regulation reservoirs, inter-basin transfers, afforestation • Conservation: low-water appliances, xeriscaping, drip irrigation, grey water, leak reduction, metering; pollution control to protect sources • Treatment: matching quality to use; screens, sedimentation, aeration, flocculation, filtration, sterilisation, pH adjustment, activated carbon, reverse osmosis • Evaluation of costs, energy, environmental impacts (habitat loss, methane, inter-tidal habitats, conflicts) and justified conclusion Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 114-127; AQA spec 3.2.2.4-3.2.2.5.	25